

Module: `Problem 2 - Printing Out the User's Guess`

This module contains a utility function used in Hangman-style games to display the current state of a secret word based on the letters the player has guessed so far. The function returns a string where correctly guessed letters are shown in their original positions and all other characters are represented by underscores.

Functions

`getGuessedWord(secretWord, lettersGuessed)`

Description

Constructs a visual representation of `secretWord` by revealing the letters that appear in `lettersGuessed` and replacing all other characters with underscores (`'_'`). The order of characters in the returned string matches the order in `secretWord` .

Parameters

- `secretWord` (*str*): The word that the user is trying to guess.
- `lettersGuessed` (*list of str*): A collection of letters that have been guessed so far.
Typical usage expects each element to be a single-character string (e.g., `['a', 'e', 'i']`).

Returns

`result` (*str*): A string composed of the correctly guessed letters and underscores for the remaining letters. The length of the string equals `len(secretWord)` .

Examples

```
# Example 1 - no guessed letters match
result = getGuessedWord('hello', ['apple', 'banana', 'cherry'])
# → '_____'

# Example 2 - one matching letter ('e')
result = getGuessedWord('test', ['a', 'b', 'c', 'd', 'e'])
# → '_e__'

# Example 3 - empty secret word
result = getGuessedWord('', [])
# → ''

# Example 4 - guessed list contains a multi-character string; no match
result = getGuessedWord('a', ['hello'])
# → '__'

# Example 5 - none of the guessed strings match any character
result = getGuessedWord('abcdef', ['test', 'word', 'example', 'data', 'value'])
# → '_____'
```

Edge Cases / Notes - The function does **not** validate that items in `lettersGuessed` are single characters. If the list contains multi-character strings (as shown in the examples), the membership test `1 in lettersGuessed` will

fail, resulting in an underscore for that position. - An empty `secretWord` returns an empty string (`''`). - The function performs a simple linear scan; its time complexity is $O(n \times m)$ where n is the length of `secretWord` and m is the length of `lettersGuessed` . For typical Hangman usage (small word lengths and short guess lists) this is negligible.